Indian Institute of Technology Hyderabad
Signal Processing, Fall 2026
Quiz 2, 31.08.2026, 10 points

1. Answer the following questions about discrete-time systems. Justify your answers.
   (a) Given an example of a discrete-time system that is *non-linear, non-causal, time-invariant, BIBO stable*. (1)
   (b) Given an example of a discrete-time system that is *linear, non-causal, not time-invariant, BIBO stable*. (1)
2. How is the $z-$transform of the output $y[n]$ of a discrete-time LTI system related to the $z-$transform of the input $x[n]$ and the system's impulse response $h[n]$? Show your work and clearly state any assumptions. (2)
3. A discrete-time LTI system computes the average of the *current* sample and the *sample before its previous* sample.
   (a) Write the expression for the system's impulse response $h[n]$. (1)
   (b) Let $x[n] = \cos(\pi n)$, and $y[n] = T\{x[n]\} = x[n] * h[n]$. Find $y[-1]$ and $y[2]$ using the graphical method. (2)
4. Let $h_{\text{diff}}[n] = 0.5(\delta[n] - \delta[n-1])$, $h_{\text{ave}}[n] = 0.5(\delta[n] + \delta[n-1])$, $h_{\text{cascade}}[n] = h_{\text{diff}}[n] * h_{\text{ave}}[n]$, where $*$ is the convolution operator. Analyze the frequency response of all three LTI systems using their DTFT magnitude spectra. What can you infer from the cascaded system's spectrum? (3)